CNENGR202 Instrumentation and Control

Part 1 L7 – Feedback Control – Algorithms II

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Outline

- Proportional Action
- Integral Action
- Derivative Feedback
- Combined Control Algorithms

Proportional Action - 1

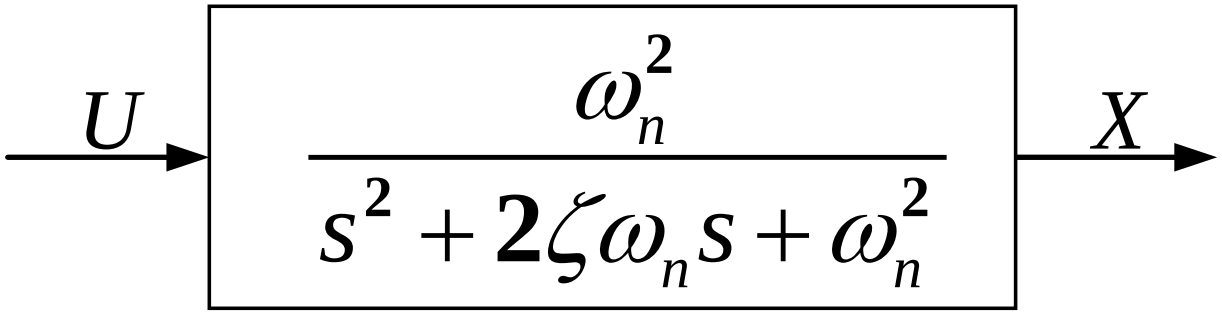
- For tutorial purposes, each algorithm is applied to a generalized 2nd order differential equation, i.e. the mathematical model of the engineering device is given by

$$\frac{d^2x}{dt^2} + 2\zeta\omega_n \frac{dx}{dt} + \omega_n^2 x = \omega_n^2 u(t)$$

$$X = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} U$$

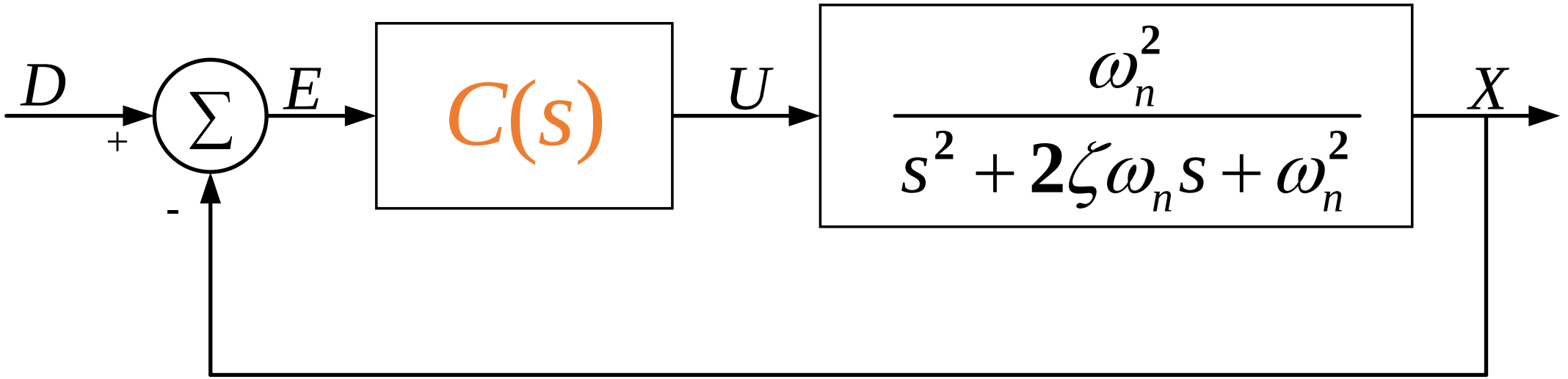
Proportional Action - 2

- Open loop block diagram

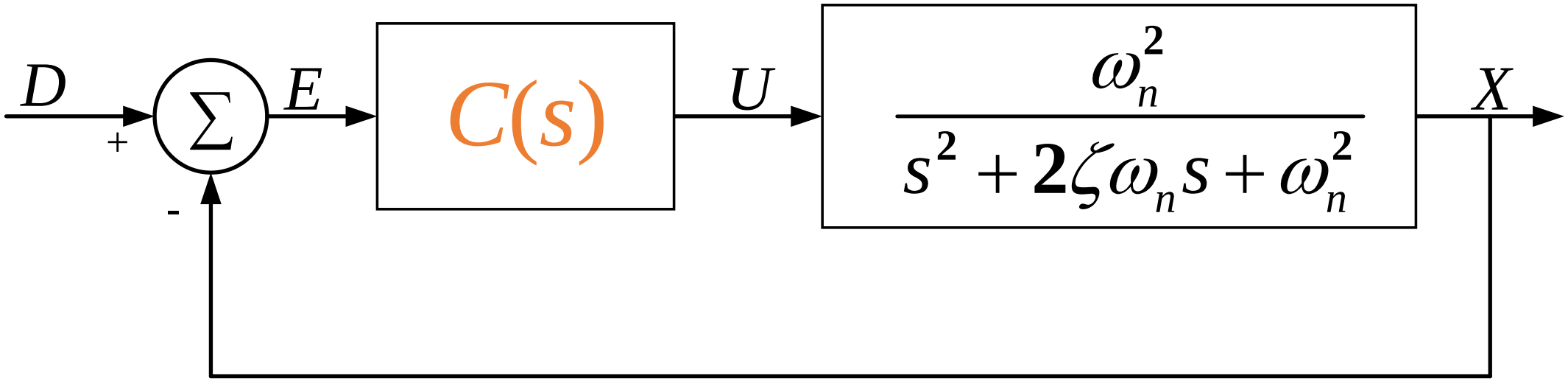


Proportional Action - 3

- Closed loop block diagram

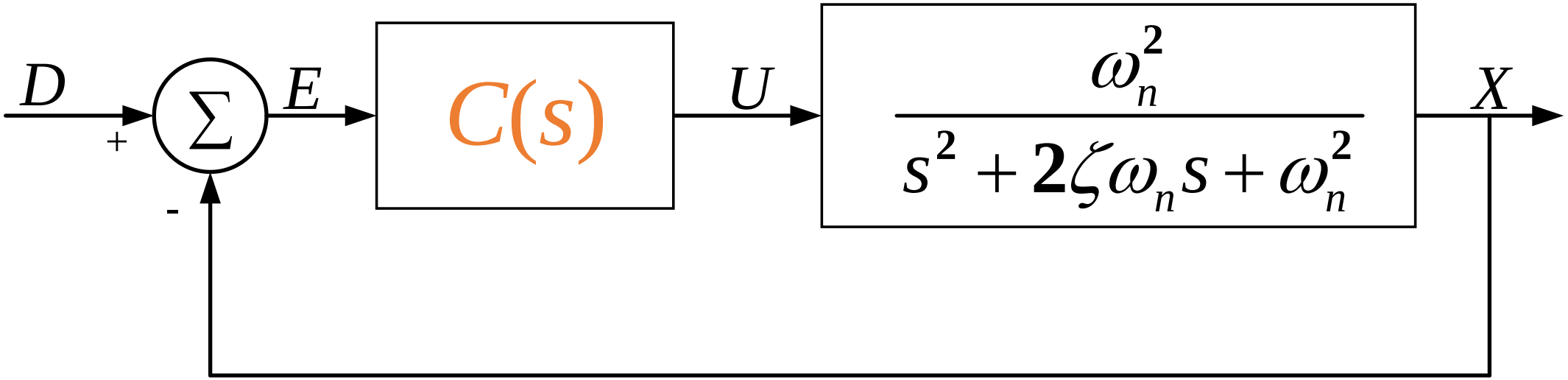


Proportional Action - 4

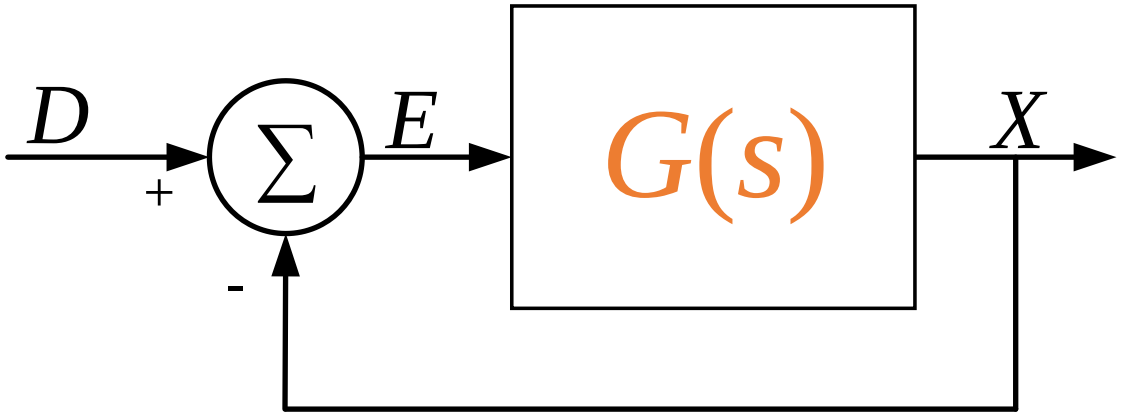


Control Law: $u(t) = k_p \times \left(d(t) - x(t)\right)$

Proportional Action - 5



$$u(t) = k_p \times \left(d(t) - x(t) \right)$$



- Find $G(s)$?
- Find X/D=?

Proportional Action - 6

- Using the Series rule $$G(s) = \frac{k_p \times \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- Then using the Negative Feedback rule

$$\frac{X}{D} = \frac{\dfrac{k_p \omega_n^{\,2}}{s^2 + 2\zeta\omega_n s + \omega_n^2}}{1 + \dfrac{k_p \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}} = \frac{k_p \omega_n^2}{s^2 + 2\zeta\omega_n s + (1 + k_p)\omega_n^2}$$

Proportional Action - 7

- Closed-Loop Transfer Function

$$X = \frac{k_p \omega_n^2}{s^2 + 2\zeta\omega_n s + \left(1 + k_p\right)\omega_n^2} D$$

- Characteristic Equation (Closed-Loop)

$$s^2 + 2\zeta\omega_n s + \left(1 + k_p\right)\omega_n^2 = 0$$

- Characteristic Equation (Plant)

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

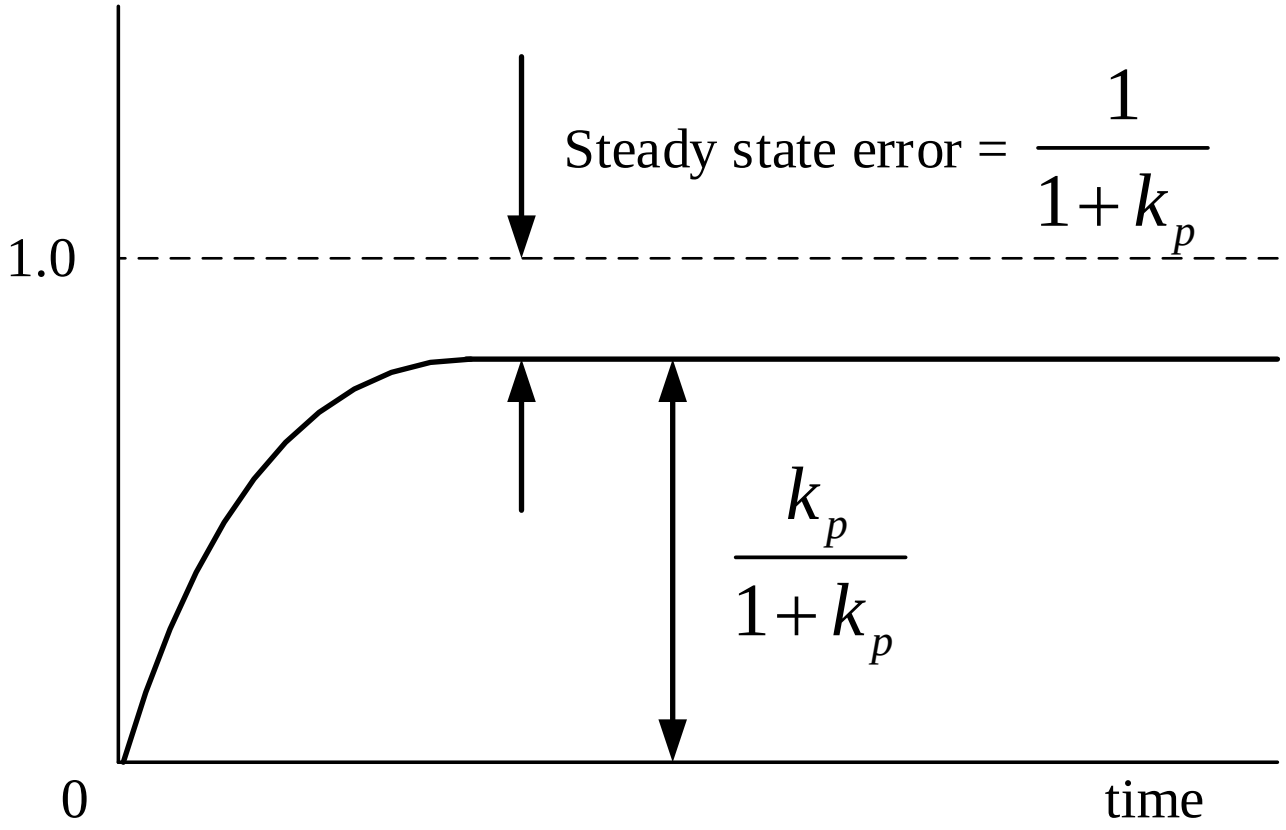
Determine the steady state behavior - 1

$$X = \frac{k_p \omega_n^2}{s^2 + 2\zeta\omega_n s + \left(1 + k_p\right)\omega_n^2} D$$

- For a unit step command input

$$x\left(t \to \infty\right) = \lim_{s \to 0} sX\left(s\right) = \lim_{s \to 0}\left[s \cdot \frac{k_p \omega_n^2}{s^2 + 2\zeta\omega_n s + \left(1 + k_p\right)\omega_n^2} \cdot \frac{1}{s} \right] = \frac{k_p}{1 + k_p}$$

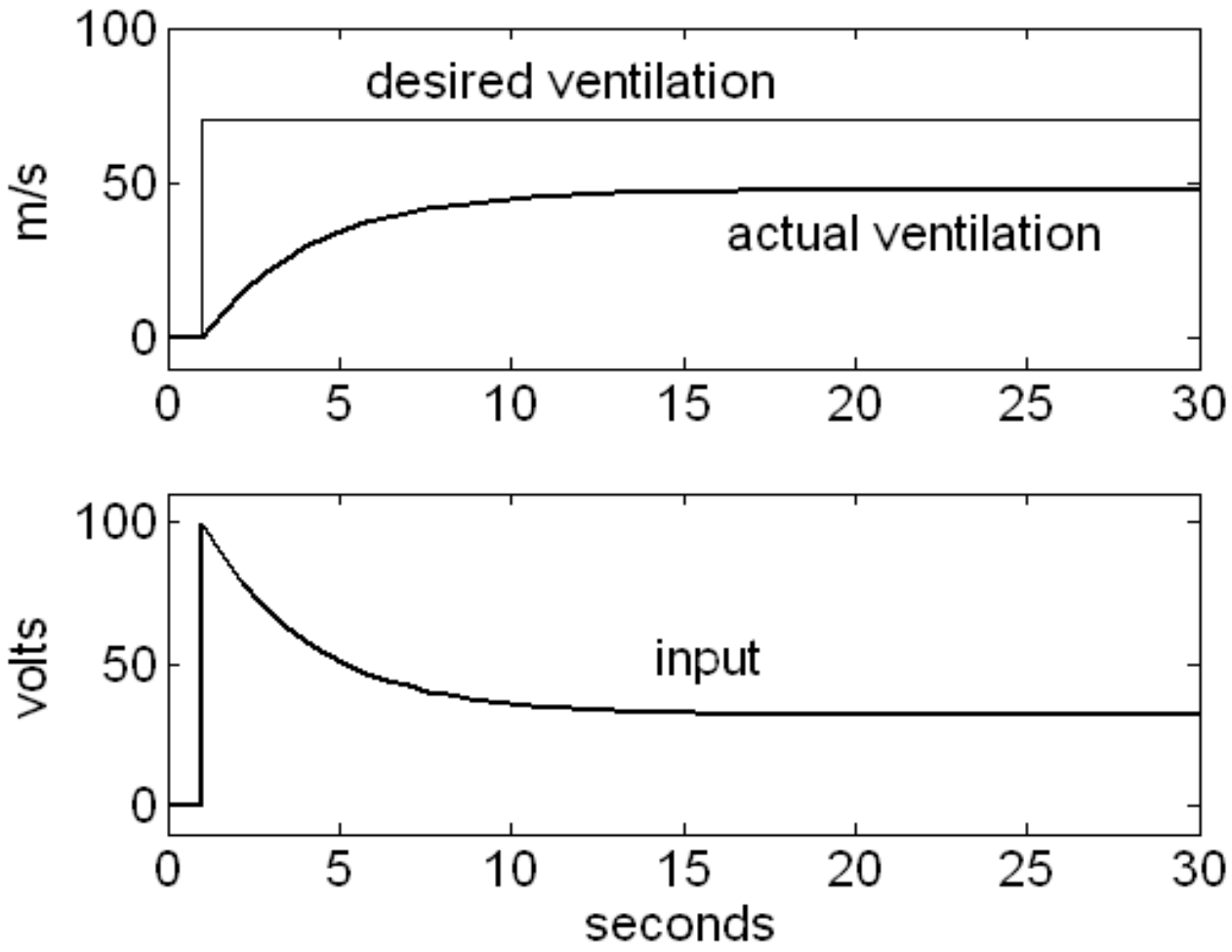
Determine the steady state behavior - 2



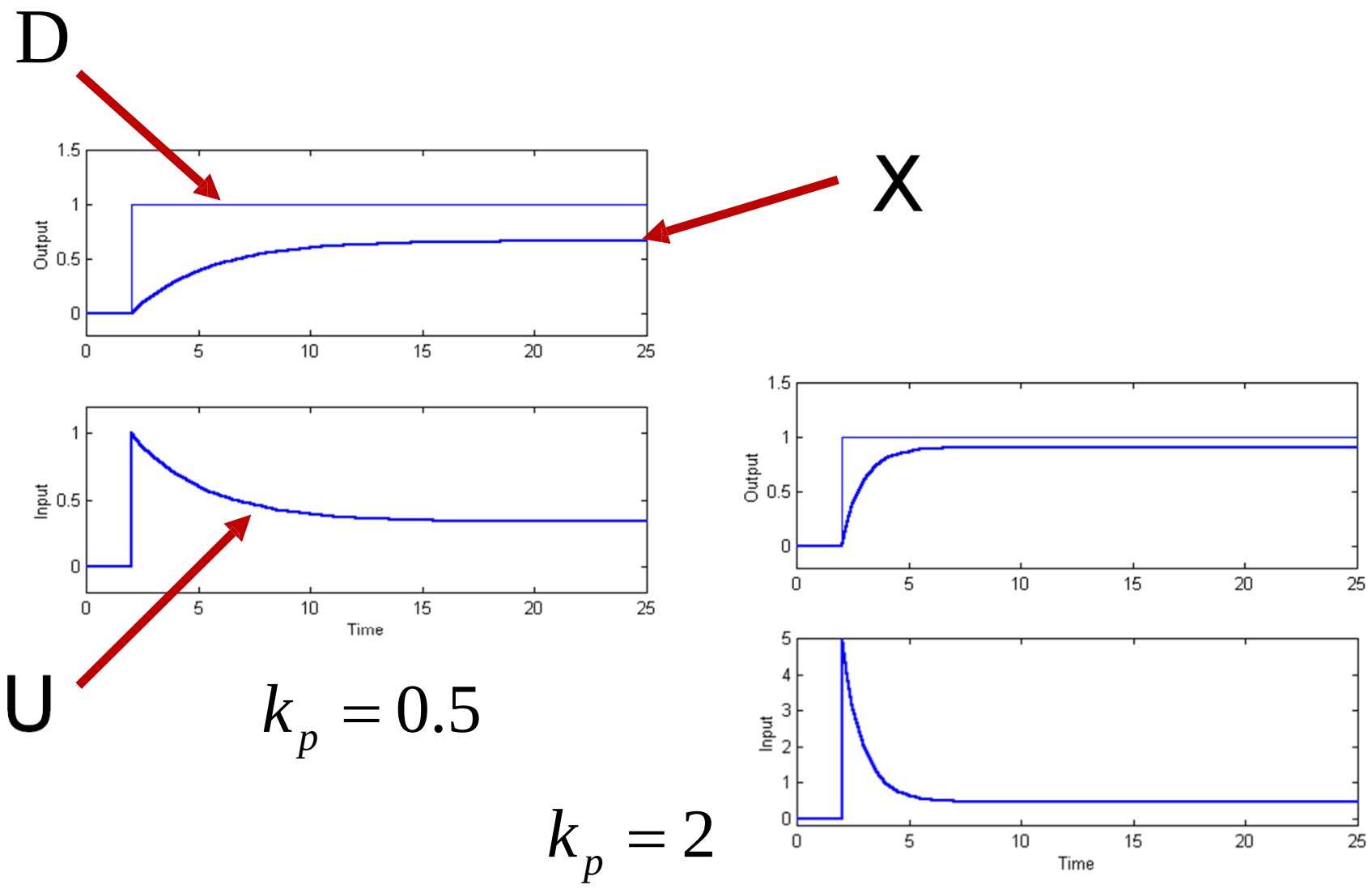
Determine the steady state behavior - 3

- Desired output or set point for this example 1.0
  - Value of the output at steady state $\frac{k_p}{1+k_p}$
  - Therefore, the steady state error is $1-\frac{k_p}{1+k_p}=\frac{1}{1+k_p}$

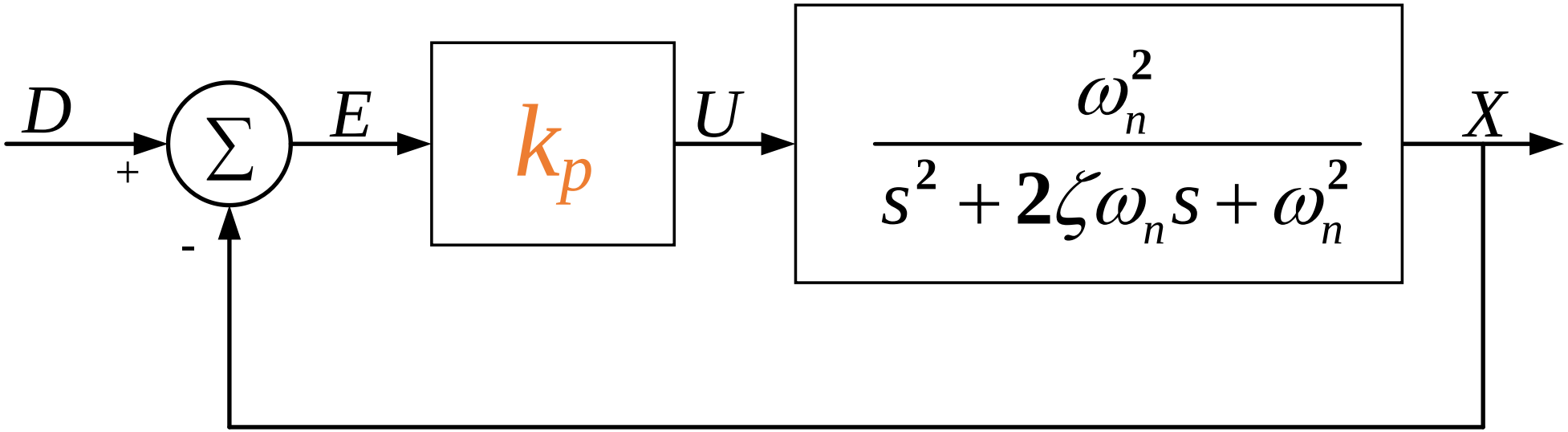
Proportional Control of Ventilation Rate using an axial fan - 1



Proportional Control of Ventilation Rate using an axial fan - 2



Example: Proportional Action - 1



Control parameters: k_p=1, ζ=2, ω_n=2, find the characteristic equation?
If D is a unit step, find the steady state value?

Example: Proportional Action - 2

- Closed-Loop Transfer Function

$$X = \frac{4}{s^2 + 8 + 8} D$$

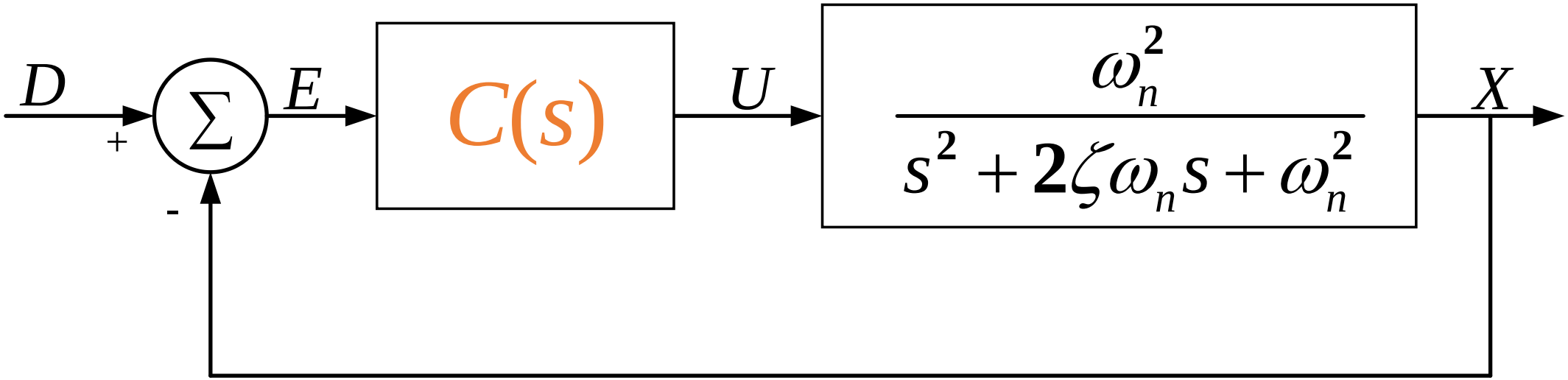
- Characteristic Equation (Closed-Loop)

$$s^2 + 8s + 8 = 0$$

- For a unit step command input

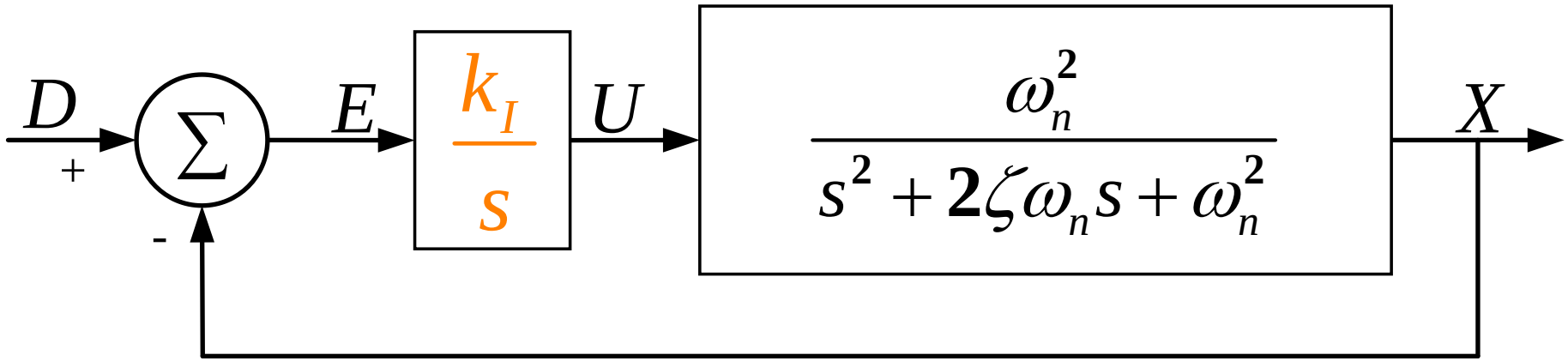
$$x\left(t \to \infty\right) = \lim_{s \to 0} sX\left(s\right) = \lim_{s \to 0} \left[s \cdot \frac{4}{s^2 + 8s + 8} \cdot \frac{1}{s} \right] = \frac{4}{8} = 0.5$$

Integral Action – 1



$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Integral Action – 2



Control Law: $$u(t) = k_I \times \int \left(d(t) - x(t) \right) dt$$

Integral Action – 3

- Closed-Loop Transfer Function:

$$X(s) = \frac{\omega_n^2 k_I}{s^3 + 2\zeta\omega_n s^2 + \omega_n^2 s + \omega_n^2 k_I} D(s)$$

- Characteristic Equation (Closed-Loop)

$$s^3 + 2\zeta\omega_n s^2 + \omega_n^2 s + \omega_n^2 k_I = 0$$

- Characteristic Equation (Plant)

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Integral Action – 4

- Determine the steady state behaviour

$$X(s) = \frac{\omega_n^2 k_I}{s^3 + 2\zeta\omega_n s^2 + \omega_n^2 s + \omega_n^2 k_I} D(s)$$

- For a unit step command input

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s) = \lim_{s\to 0}\left[s \cdot \frac{\omega_n^2 k_I}{s^3 + 2\zeta\omega_n s^2 + \omega_n^2 s + \omega_n^2 k_I} \cdot \frac{1}{s} \right] = 1.0$$

Integral Action – 5

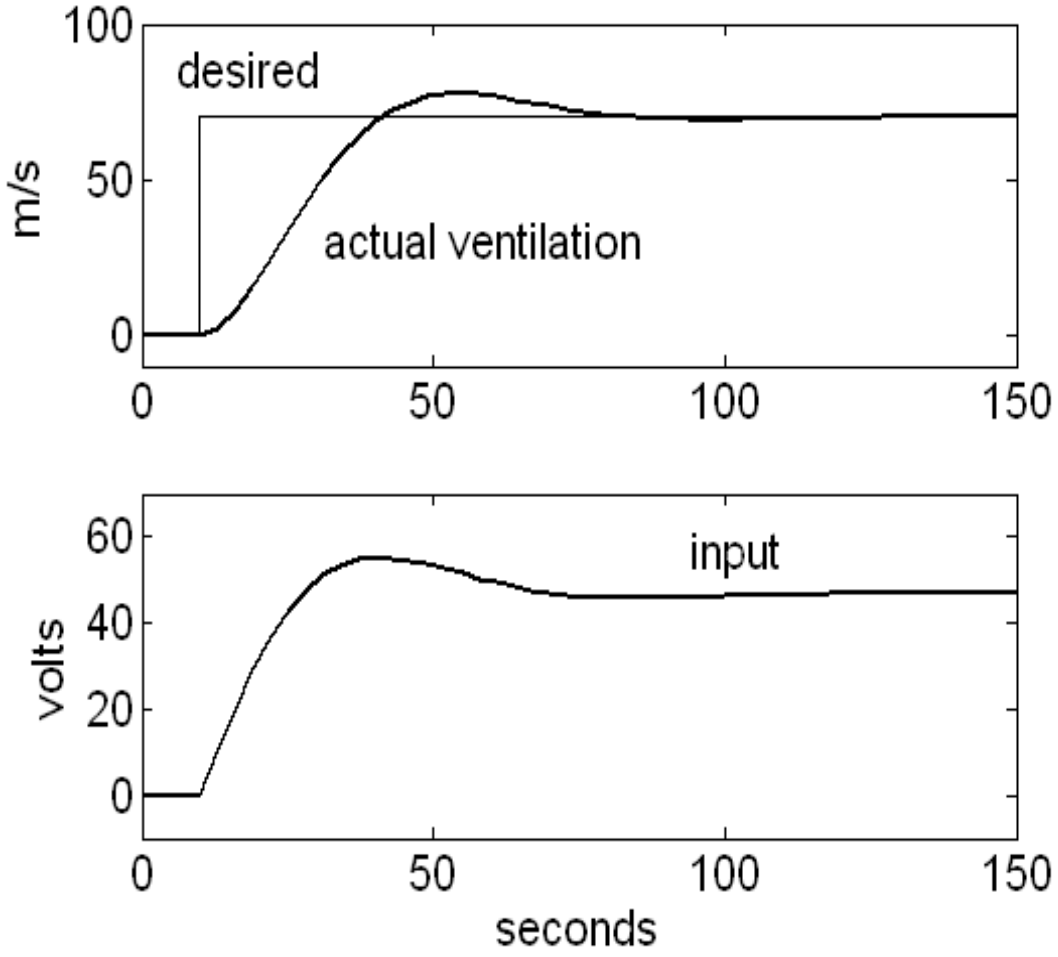
- The steady state gain of the closed-loop Transfer Function is unity:

$$x\left(t \to \infty\right) = 1.0$$

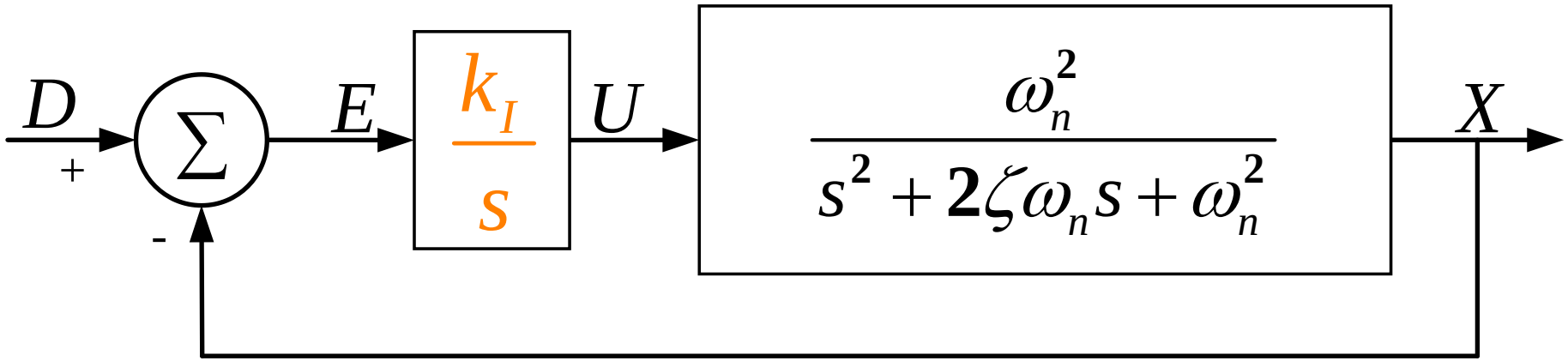
- There is steady state tracking of the set point, i.e. the output asymptotically approaches the desired output value!

Integral Action – 6

Integral Control of Ventilation Rate using an axial fan



Example: Integral Action - 1



Control parameters: k_I=2 , ζ=2, ω_n=2, find the characteristic equation?

If *D* is a unit step, find the steady state value?

Example: Integral Action - 2

- Closed-Loop Transfer Function

$$X(s) = \frac{8}{s^3 + 8s^2 + 4s + 8} D(s)$$

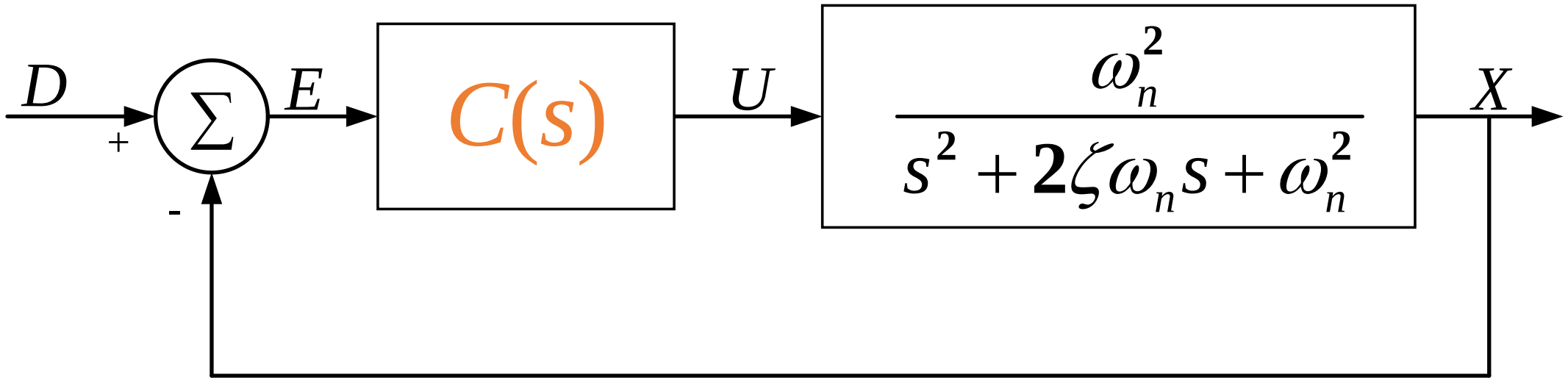
- Characteristic Equation

$$s^3 + 8s^2 + 4s + 8 = 0$$

- Steady state for a unit step command input

$$\lim_{t \to \infty} x(t) = \lim_{s \to 0} sX(s) = \lim_{s \to 0} \left[s \cdot \frac{8}{s^3 + 8s^2 + 4s + 8} \cdot \frac{1}{s} \right] = 1.0$$

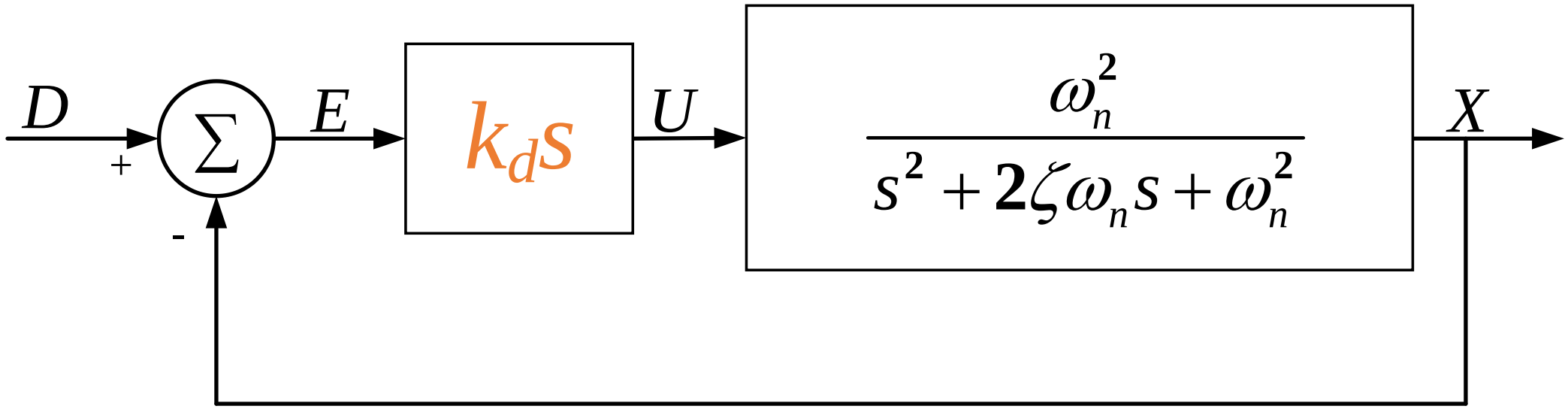
Derivative Feedback - 1



$$u(t) = k_d \frac{de(t)}{dt} = k_d \frac{d}{dt}\left(d(t) - x(t)\right)$$

$$U = k_d s\left(D - X\right)$$

Derivative Feedback - 2



Control Law
Transfer Function form)

$$U = k_d s\left(D - X\right)$$

Derivative Feedback - 3

$$X(s) = \frac{\dfrac{k_d \omega_n^2 s}{s^2 + 2\zeta\omega_n s + \omega_n^2}}{1 + \dfrac{k_d \omega_n^2 s}{s^2 + 2\zeta\omega_n s + \omega_n^2}} D(s)$$

$$= \frac{k_d \omega_n^2 s}{s^2 + \left(2\zeta\omega_n + k_d \omega_n^2\right) s + \omega_n^2} D(s)$$

Derivative Feedback - 4

- Closed-Loop Transfer Function

$$X(s) = \frac{k_d \omega_n^2 s}{s^2 + \left(2\zeta\omega_n + k_d\omega_n^2\right)s + \omega_n^2} D(s)$$

- Characteristic Equation (Closed-Loop)

$$s^2 + \left(2\zeta\omega_n + k_d\omega_n^2\right)s + \omega_n^2 = 0$$

- Characteristic Equation (Plant)

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Determine the steady state behavior

$$X(s) = \frac{k_d \omega_n^2 s}{s^2 + \left(2\zeta\omega_n + k_d\omega_n^2\right)s + \omega_n^2} D(s)$$

- For a unit step command input

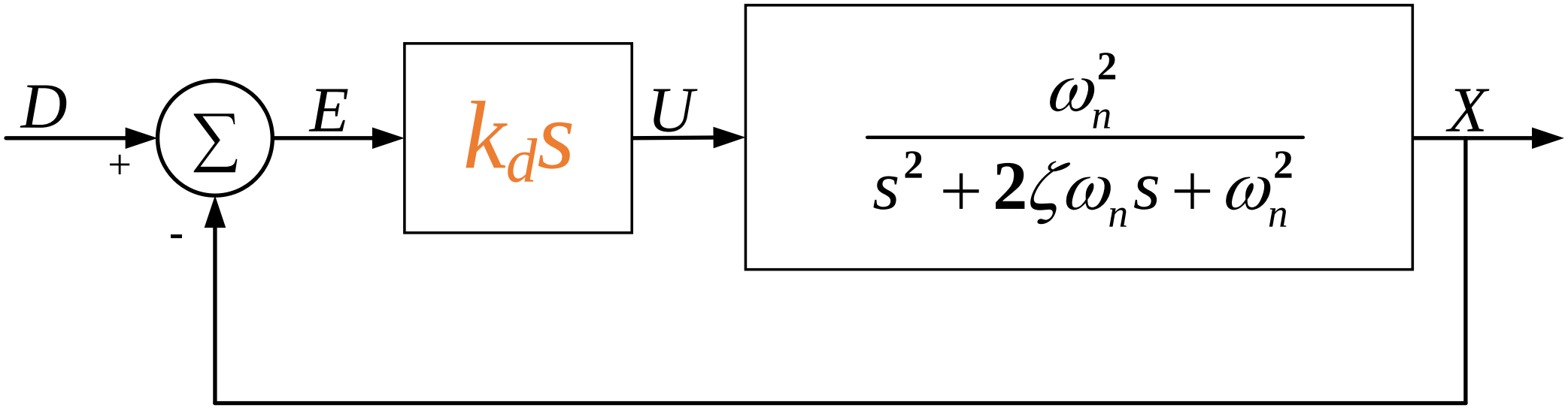
$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s) = \lim_{s\to 0}\left[s \cdot \frac{\omega_n^2 k_d s}{s^2 + \left(2\zeta\omega_n + \omega_n^2 k_d\right)s + \omega_n^2} \cdot \frac{1}{s} \right] = 0$$

Time domain

- Derivative Action (time domain)

$$u(t) = k_d \frac{de(t)}{dt} = k_d \frac{d}{dt}\left(d(t) - x(t)\right)$$

Example: Derivative Feedback - 1



Control parameters: k_d=5, ζ=2, ω_n=2, find the characteristic equation?

If D is a unit step, find the steady state value?

Example: Derivative Action - 2

- Closed-Loop Transfer Function

$$X(s) = \frac{20s}{s^2 + 28s + 4} D(s)$$

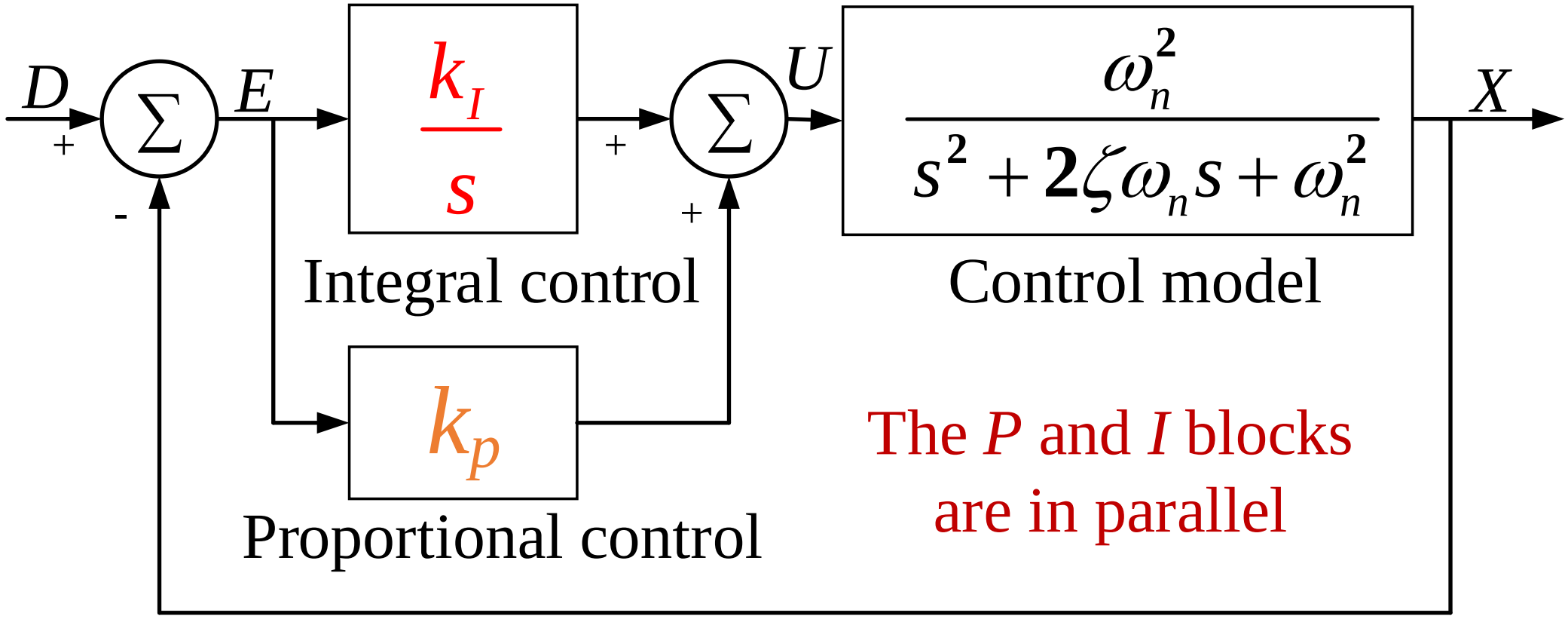
- Characteristic Equation

$$s^2 + 28s + 4 = 0$$

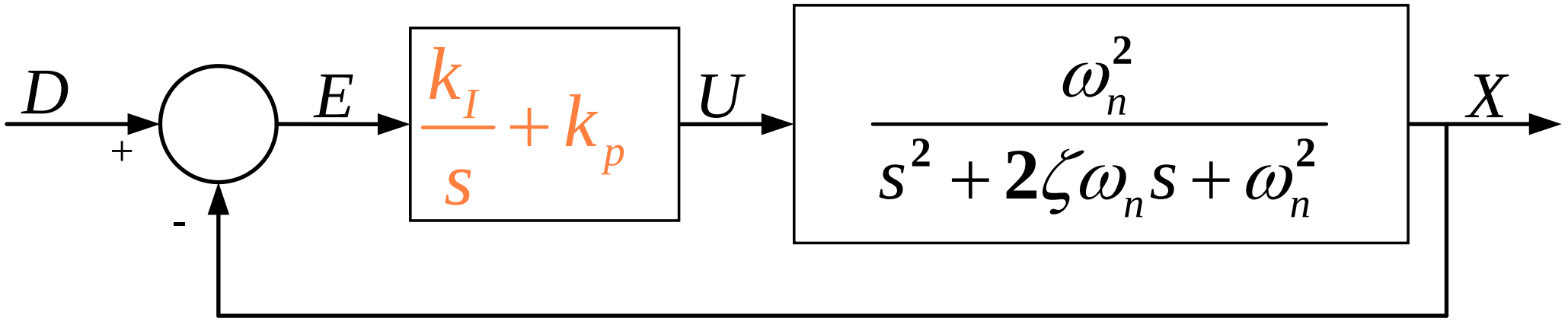
- Steady state for a unit step command input

$$\lim_{t \to \infty} x(t) = \lim_{s \to 0} sX(s) = \lim_{s \to 0} \left[s \cdot \frac{20s}{s^2 + 28s + 4} \cdot \frac{1}{s} \right] = 0$$

PI Control - 1


D
+
-
E
k_I
s
Integral control
+
+
U
ω_n^2
$s^2 + 2\zeta\omega_n s + \omega_n^2$
Control model
X
k_p
Proportional control
The P and I blocks
are in parallel

PI Control - 2



Control Law: $u(t) = k_p e(t) + k_I \int e(t)\,dt$

PI Control - 3

- Closed-Loop Transfer Function

$$X(s) = \frac{\omega_n^2 \left(k_I + k_p s\right)}{s^3 + 2\zeta\omega_n s^2 + \left(1 + k_p\right)\omega_n^2 s + k_I \omega_n^2} D(s)$$

- Characteristic Equation (Closed-Loop)

$$s^3 + 2\zeta\omega_n s^2 + \left(1 + k_p\right)\omega_n^2 s + k_I \omega_n^2 = 0$$

- Characteristic Equation (Plant)

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

PI Control - 4

- Determine the steady state behavior

$$X(s) = \frac{\omega_n^2 \left(k_I + k_p s\right)}{s^3 + 2\zeta\omega_n s^2 + \left(1 + k_p\right)\omega_n^2 s + k_I \omega_n^2} D(s)$$

- For a unit step command input

$$x\left(t \to \infty\right) = \lim_{s \to 0} sX(s) = \lim_{s \to 0} \left[s \frac{\omega_n^2 \left(k_I + k_p s\right)}{s^3 + 2\zeta\omega_n s^2 + \left(1 + k_p\right)\omega_n^2 s + k_I \omega_n^2} \frac{1}{s} \right] = 1.0$$

- The steady state gain is unity…

PI Control - 5

- The steady state gain of the closed-loop Transfer Function is unity

$$x\left(t \to \infty\right) = 1.0$$

- There is steady state tracking of the set point, i.e. the output asymptotically approaches the desired output value!

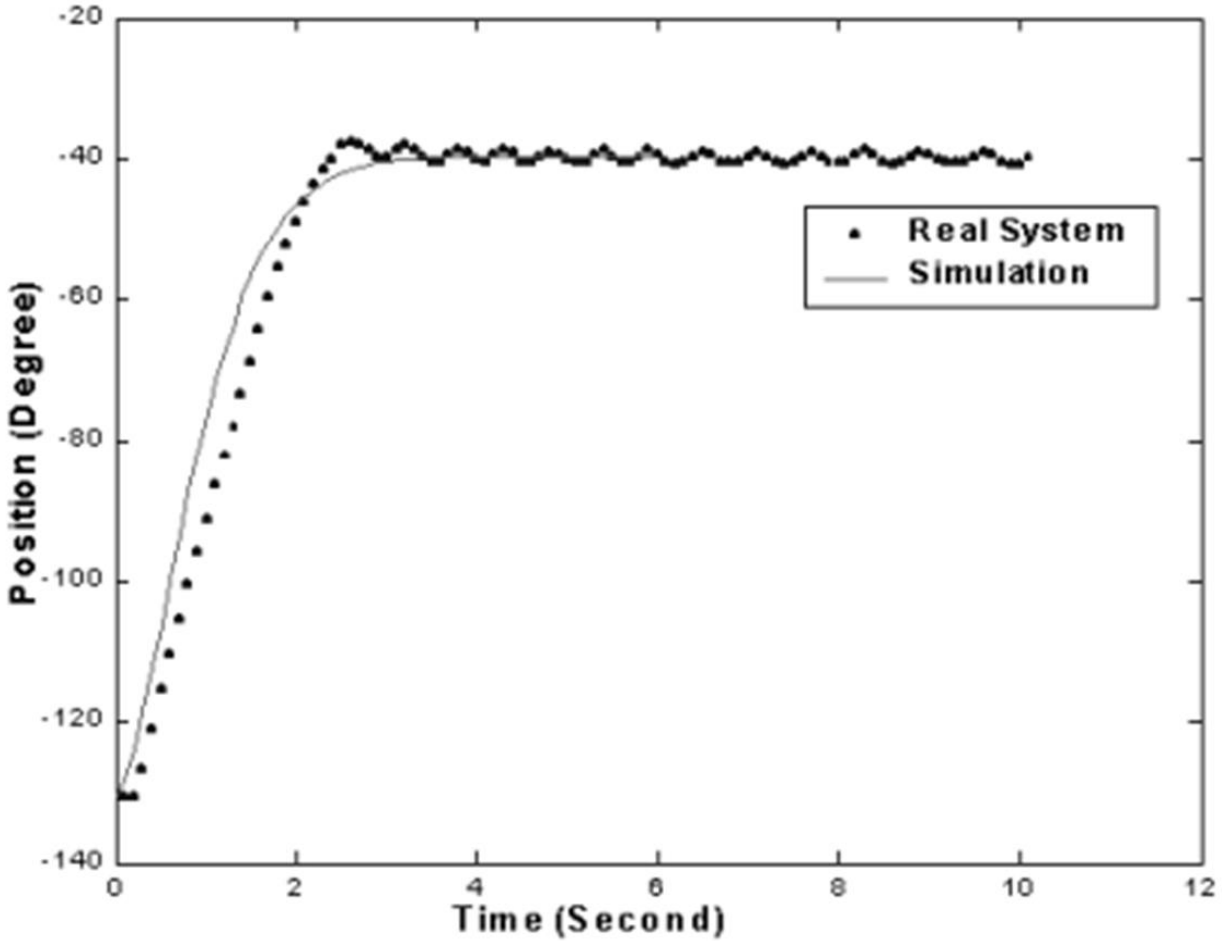
PI Control - 6

A Computerised Intelligent Excavator, developed for the excavation of foundations and general earthworks



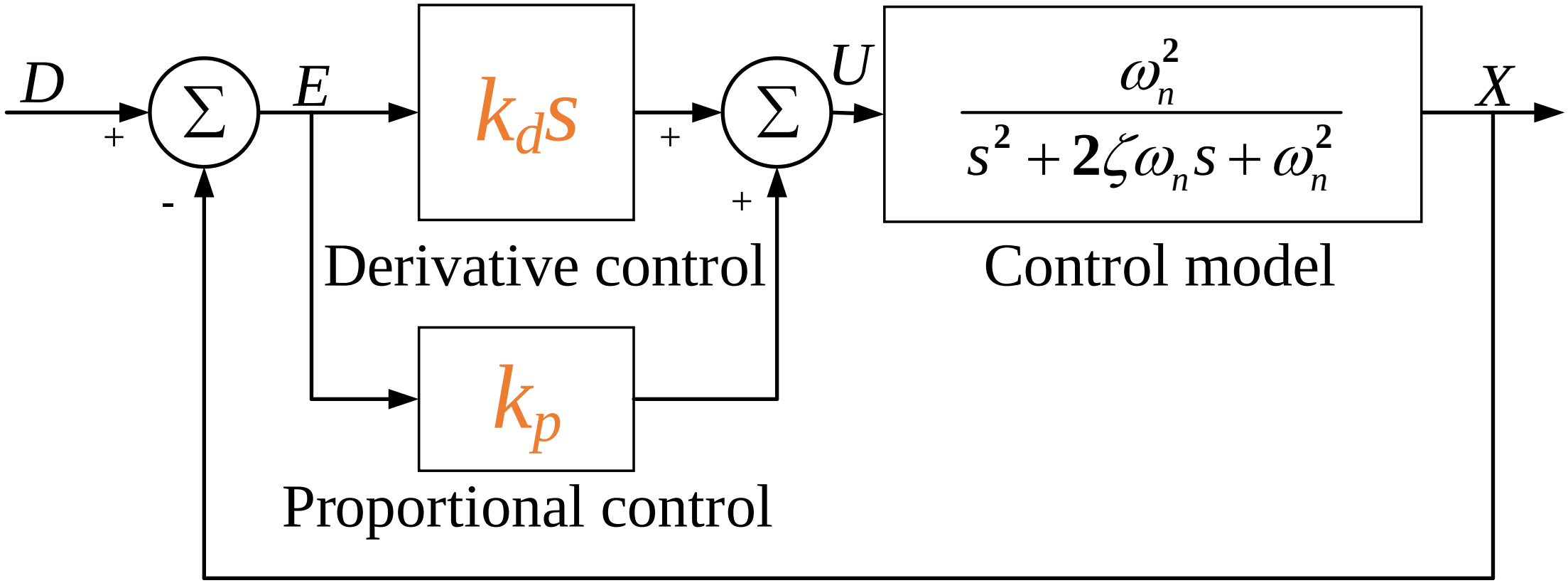
PI Control - 7

Comparison of simulation response and real data, showing problematic oscillations – additional control element required



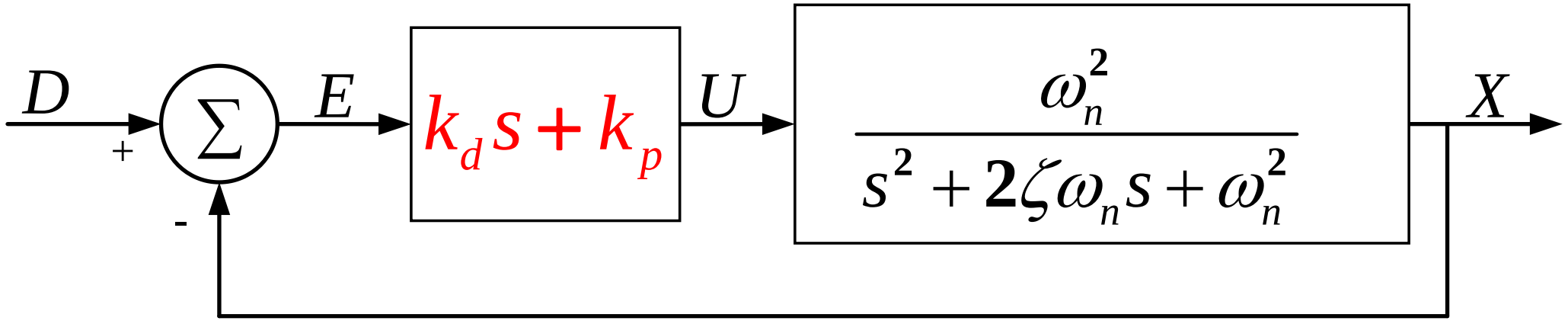
PI Control

PD Control - 1



$$u(t) = k_p e(t) + k_d \frac{de(t)}{dt}$$

PD Control - 2



PD Control - 3

- Closed-Loop Transfer Function

$$X(s) = \frac{\omega_n^2 \left(k_p + k_d s\right)}{s^2 + \left(2\zeta\omega_n + k_d\omega_n^2\right)s + \left(1 + k_p\right)\omega_n^2} D(s)$$

- Characteristic Equation (Closed-Loop)

$$s^2 + \left(2\zeta\omega_n + k_d\omega_n^2\right)s + \left(1 + k_p\right)\omega_n^2 = 0$$

- Characteristic Equation (Plant)

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

- The 'stiffness' (speed) and damping can be manipulated using the two control gains

PD Control - 4

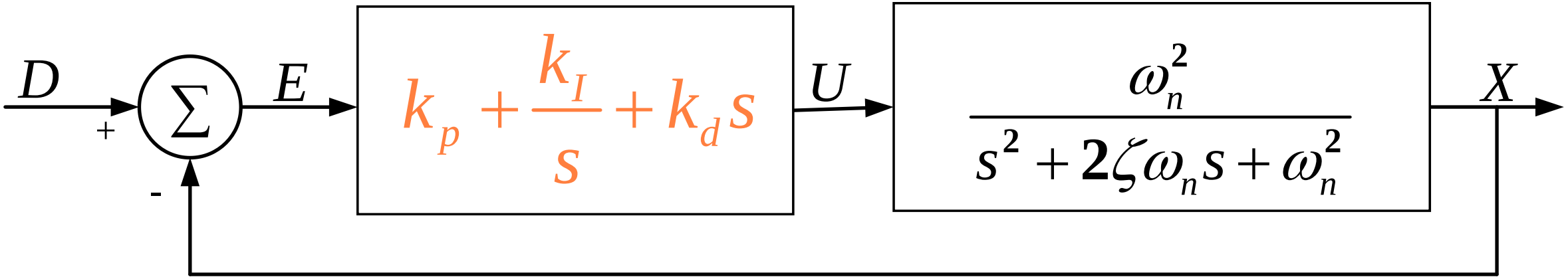
- Determine the steady state behavior

$$X(s) = \frac{\omega_n^2 \left(k_p + s k_d \right)}{s^2 + \left(2\zeta\omega_n + \omega_n^2 k_d \right) s + \omega_n^2 \left(1 + k_p \right)} D(s)$$

- For a unit step command input

$$\lim_{t \to \infty} x(t) = \lim_{s \to 0} sX(s) = \frac{k_p}{1 + k_p}$$

PID Control - 1



Control Law:

$$u(t) = k_p e(t) + k_I \int e(t)\,dt + k_d \frac{de(t)}{dt}$$

PID Control - 2

- Closed-Loop Transfer Function

$$X(s) = \frac{\omega_n^2 \left(k_d s^2 + k_p s + k_I \right)}{s^3 + \left(2\zeta\omega_n + \omega_n^2 k_d \right) s^2 + \left(1 + k_p \right) \omega_n^2 s + k_I \omega_n^2} D(s)$$

- Characteristic Equation (Closed-Loop)

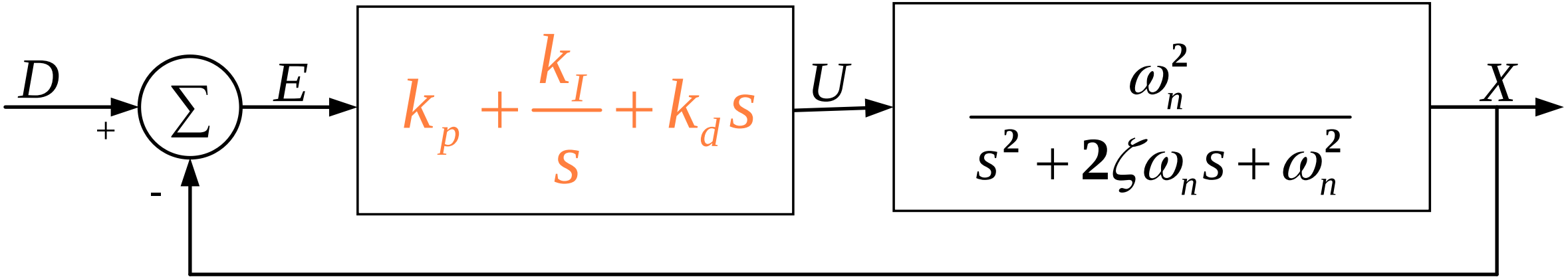
$$s^3 + \left(2\zeta\omega_n + \omega_n^2 k_d \right) s^2 + \left(1 + k_p \right) \omega_n^2 s + k_I \omega_n^2 = 0$$

- Characteristic Equation (Plant)

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

- When PID is applied to a second order model, there are three poles and three gains…

Example: PID Control - 1



Control parameters: k_p=1, k_I=2, k_d=5, ζ=2, ω_n=2, find the characteristic equation?

If D is a unit step, find the steady state value?

Example: PID Control - 2

- Closed-Loop Transfer Function

$$X(s) = \frac{20s^2 + 4s + 8}{s^3 + 28s^2 + 8s + 8} D(s)$$

- Characteristic Equation

$$s^3 + 28s^2 + 8s + 8 = 0$$

- Steady state for a unit step command input

$$x(\infty) = \lim_{s \to 0} sX(s) = \lim_{s \to 0} \left(s \cdot \frac{20s^2 + 4s + 8}{s^3 + 28s^2 + 8\omega_n{}^2 s + 8} \cdot \frac{1}{s} \right) = 1$$

Combined Control Algorithms

Which Control Law to use ?

- PD Control: Useful in cases when a steady state error is tolerated; can use to increase damping
- PI Control: Mainly useful for first order systems or when we have relaxed performance requirements
- PID Control: For 2nd order or higher systems
- More advanced controllers: Required for ‘difficult’ high order and/or nonlinear systems
- Tuning Methods: trial and error; frequency domain methods; root locus methods; model-based methods